

Post-detection related developments, and SETI in Society news: 2022.

(PC Members: please see full written report for details, links, etc.)

An update for the IAA SETI Permanent Committee, 19 Sept 2022, at IAC Paris

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Overview of what will be covered:

- Current status: since last report in 2021
- Papers on themes related to Post-detection in 2022 IACs
- Some relevant recent and upcoming activities of SETI PC members
- Some conferences and symposia with post-detection relevance
- Some relevant research centres and seminar series (established and emerging)
- A few noteworthy publications
- A 'SETI in the Media' Case Study for 2022: FAST, June 2022
- Update from John Elliott and Martin Dominik on the UKSRN Post-Detection Hub (St. Andrews)
- General Q&A

General Status

- We have many members interested in post-detection, but we aren't currently formally organized into a working group with a specific remit. Can be revisited when ToR are finalized / current membership is clear.
- The last update in 2021 suggested some specific ways forward which are still applicable (work towards timeless Principles + more dynamic Best Practices / Codes of Conduct). N.b. we need PD background research to inform revised policy. And we need funding for that.
- Meanwhile, considerable research and discussion relevant to PD does continue
- Outbound transmissions from Earth (de novo or reply) still require constructive discussion
- To watch: SoE for biosignatures / confidence in life detection discussions; newer research on escalating mis/disinformation online (an extremely serious problem)

1. Papers related to Post-detection: IAC 2022

<https://iafastro.directory/iac/browse/IAC-22/A4/>

- Daniela De Paulis, A Sign in Space: an interdisciplinary exploration of the potential reception of an extraterrestrial signal. <https://iafastro.directory/iac/paper/id/67928/summary/> (Billingham Cutting-Edge Lecture)
- Claudio Maccone, Fifty Years of SETI in the IAF Digital Library. <https://iafastro.directory/iac/paper/id/73236/summary/>
- Ugur Guven, Spaceships in SETI Research and Potential for Spaceflight Technology. <https://iafastro.directory/iac/paper/id/67299/summary/>
- Suresh Bhattari and Manisha Dwa, Are we alone? People's perception on an ExoWorld, its planet and possible life on it. <https://iafastro.directory/iac/paper/id/71922/summary/>
- Mclee Kerolle, Ridha Aditya Nugraha, Sebastian Hernandez, Pragati Mot, The Legal Implications of Belonging to an Extraterrestrial Polity. <https://iafastro.directory/iac/paper/id/73495/summary/>

1. Papers related to Post-detection: IAC 2022, cont'd

<https://iafastro.directory/iac/browse/IAC-22/A4/>

- Léanne Fortuna, The World Configuration of Aliens Were Our Neighbours. <https://iafastro.directory/iac/paper/id/69479/summary/>
- Carol Oliver, Astrobiology immersive Virtual Field Trip creation now as easy as PowerPoint. <https://iafastro.directory/iac/paper/id/71127/summary/>
- Rebecca Charbonneau, CETI Across the Iron Curtain. <https://iafastro.directory/iac/paper/id/71778/summary/>
- Gabriel G. De la Torre and Celia Avila-Rauch, Romanticism in Science as a form of cognitive bias and SETI. <https://iafastro.directory/iac/paper/id/73175/summary/>

2. Some relevant recent and upcoming activities of SETI Permanent Committee members

- N.b. This compilation is not comprehensive. Also see updates below
- **Continued public outreach by SETI Institute and Breakthrough Listen**
- **The UK SETI Research Network (UKSRN), meeting and imminent Post-Detection Hub (John Elliott and Martin Dominik will say more at the end)**
 - [The Post-Detection Hub](#), a UKSRN initiative, is on the verge of launching. Jointly hosted by the St Andrews Centre for Exoplanet Science and the Centre for Global Law and Governance. Additional affiliates to be announced.
 - The [annual conference](#) was this year organized/hosted by Michael Bohlander at Durham in July, and included papers relevant to post-detection by Michael Bohlander, Kate Genevieve, and John Elliott.
- **PSETI (Penn State) activities including:**
 - PSETI 2022 symposium, included a post-detection session involving PC members and others (more below)
 - [SETI Bibliography Project](#)
 - [SETI News](#)
 - [“SETI in 2021”](#) Helpful review by Macy Huston and Jason Wright.

2. Some relevant recent and upcoming activities of SETI Permanent Committee members, cont'd

Confidence in Life Detection (CoLD) scale and Standards of Evidence for Biosignatures

Substantial discussion has occurred over the past 18 months (perhaps mostly in America / the NASA-sphere) regarding the CoLD scale and Standards of Evidence for Biosignatures, involving some members of the technosignatures community. It sounds as though there are future workshops in the planning stages, including more interdisciplinary researchers (journalism, social media, science communication, social science, etc.) This is all highly relevant to post-detection, announcement of life detection, etc. To be continued... See [current talks in Goddard Technosignatures series](#).

- Green, J. et al 2021. [Call for a framework for reporting evidence for life beyond Earth](#). Nature 598: 575–579.
- Meadows, V., H. Graham et al. 2022. [Community Report from the Biosignatures Standards of Evidence Workshop](#).
- [NExSS & NfoLD Standards of Evidence Joint Community Workshop description and talks](#).
- NASA Committee on Astrobiology and Planetary Science, July 2022: [Independent Review of the Community Report from the Biosignature Standards of Evidence Workshop](#)
- N.b. so far, there has been minimal input from specialists in the social sciences, history of science, science communication research, etc. For more on the interdisciplinary side of this, see K. Denning's [brief AbSciCon 2022 talk](#), and the resources from the PSETI Post-detection breakout session (see below).

2. Some relevant recent and upcoming activities of SETI Permanent Committee members, cont'd

Daniela de Paulis and team, A Sign in Space

- This project, presented by PC member de Paulis during last year's PC meeting and again this year in the SETI sessions as the Billingham Cutting Edge Lecture, involves considerable discussion with an advisory group regarding post-detection issues and the public. De Paulis is also a [SETI Institute Artist in Residence](#).

3. Some Conferences and Symposia with post-detection relevance

- **The First Penn State SETI Symposium (June 27–30, 2022, State College, PA)** involved some members of the PC (planning, speaking, and attending), and included a variety of talks, mostly technical, but also Rebecca Charbonneau on “Review: Social Sciences and SETI”, Hilding Neilson on “The Drake Equation and the Search for Life through Indigenist Lenses”, and Chelsea Haramia on “Is there a Difference Between Messaging and Mere Detectability?”, as well as some posters with post-detection relevance. [Abstracts and talk recordings are available here, and posters too.](#)
- The [Breakout sessions](#) included two interdisciplinary sessions, one specifically on Post-Detection, which some PC members attended (in person and remotely).
- [“Post-detection for the 2020s: fundamentals, current concerns, and new discussions” \(continued on next page\)](#)

3. Some Conferences and Symposia with post-detection relevance, cont'd

- [“Post-detection for the 2020s: fundamentals, current concerns, and new discussions”](#) was run as a hybrid session by Kathryn Denning, Brian McConnell, Rebecca Charbonneau, Chelsea Haramia, George Profitiotis, with additional presenters Jill Tarter, Shelley Wright. Total: six hours of talks and active discussion exercises focused on Post-Detection themes, including astronomy, social science, history, communication systems engineering, methods from philosophy and Futures Studies.
 - [Full “Post-detection for the 2020s” breakout program with abstracts](#) .
 - The Post-detection Breakout session **talk recordings will be posted soon** (probably within a week or so) to the PSETI symposium website and the Zenodo page. Email kdenning@yorku.ca for more information and exact links.
 - **A substantial Google Drive folder of post-detection resources** was created for participants. It includes many papers from the PC’s history, and some current resources. **Please use it.** PC members who would like access can [click here](#) and request it, or email kdenning@yorku.ca
 - PSETI Dates for 2023: in the June 18-23 window. Hopefully there will be another post-detection session.

3. Some Conferences and Symposia with post-detection relevance, cont'd

- **COSPAR 2022:** Session: *Quest to Detect Extraterrestrial Life: Scientific Approaches and Cultural Impact of Discovery*. Also: new COSPAR panel on Social Sciences and Humanities
- **AbSciCon 2022:** Sessions: *Scientists, Publics, Planetary Protection and Post-Detection: Urgent Discussions for the 2020s*, and *Priorities and Strategies for Technosignature Detection*
- **SSOCIA 2022:** numerous papers
- **Islamic Perspectives on Exotheology 2022**

4. Some Research Centres and Seminar Series

- **The Technosignatures Seminar Series** at NASA Goddard continues to feature speakers on a variety of technosignatures topics, including some from SETI PC members, and some directly relevant to post-detection, e.g.,
- Jim Green on 14 Sept 2022 on “The Confidence of Life Detection Scale”,
- Jill Tarter on 21 Sept 2022 (tomorrow!) on “When Is The Evidence Sufficient To Claim The Discovery of Extraterrestrial Life?”.
- [Website with past and future speaker list and recordings of presentations.](#) To join the mailing list, contact Ravi: ravikumar.kopparapu@nasa.gov

4. Some Research Centres and Seminar Series, cont'd

- A number of university centres have emerged recently / are emerging which address philosophical and/or social issues related to astrobiology and SETI. They include, for example:
- **The Post-Detection Hub (noted above):** <https://seti.wp.st-andrews.ac.uk/>
- [Exploring Uncertainty and Risk in Contemporary Astrobiology \(EURiCA\)](#), Durham University, UK, at the Centre for Humanities Engaging Science and Society (CHESS), including a seminar series on [Philosophical Issues in Space Science](#)
- Also at Durham: **The Centre for Cosmic Impact (SETI @ Durham):** Not much information available yet, but Dr Hermine Landt-Wilman [presented on it at BIS](#) in July. John Elliott and Michael Bohlander are also connected.
- [Würzburg University. Interdisciplinary Research Centre for Extraterrestrial Studies](#), led by Hakan Kayal.

5. A few noteworthy publications

- A list of a few relevant recent papers and books is included in the report to the Committee. In addition to edited and sole-authored volumes, there is ongoing interest from many corners concerning post-detection matters, in journals including JBIS, Space Policy, IJ Astrobiology, Acta Astronautica, Am J International Law, Astrobiology, etc.
- Topics include: public interest in extraterrestrials; political considerations and possible political problems after a detection; first contact protocols; METI; legal questions about UN involvement in contact scenarios; metalaw questions concerning relationships with extraterrestrial societies; public opinion concerning hypothetical future responses to ETI, etc. Philosophy of astrobiology / ETL continues to grow. There is also an emerging discussion about decolonizing approaches to extraterrestrial life.

6. A 'SETI in the Media' Case Study for 2022: FAST: premature/confused press / social media flurry in June 2022

- (See written report for some articles / links / commentary.
- As is generally the case with potential signals, this progressed very quickly to widespread media coverage within a day.
- The range of headlines and content was significant: even in first-generation articles (let alone those derived from them), there was a wide range of descriptions of what had happened, and of the significance and implications. The accompanying illustrations varied from the classic grey alien to the FAST dish.
- Each incident like this provides valuable data about how information (and misinformation) would flow in the event of a strong candidate or confirmed detection. (But we need proper studies of this!)

And...

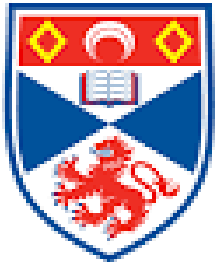
**John Elliott and Martin Dominik: UK SETI Research Network
Post-Detection Hub**



The Post Detection Hub

Official launch – Sept / Oct 2022

Years in the planning – decades of SETI involvement to conceive



University of
St Andrews

+ other affiliates to be
announced

Aim of the PD-Hub

The PD Hub's primary role will be to serve as a dedicated international 'home' for facilitating and coordinating research into a coherent post-detection framework: drawing interested SETI community colleagues together, to work on topics ranging from message decipherment and data analytics to development of regulatory protocols, space law and societal impact strategies.

Regular day-to-day operations shall include organising symposia, seminars and conferences, presentations, publications, funding applications and drafting proposals for, as well as meetings with, external, and especially with domestic and international political and governmental, stakeholders, and activities of a similar nature.



Our base to build upon:

Current UKSRN PD research includes:

- Communication Theory
- Social Media /Societal impact analytics & strategies
- Protocols
- Space Law
- Metalaw / Legal impacts on Human Civilisation
- METI implications
- LARP (Live Action Role Play)
- Signal categorisation, decoding & analytics
- Royal Society - From Afar Science exhibition <https://www.fromafar.world/> - ongoing survey on Contact
- Theological implications of Contact
-
- + much more in the domain of Astrophysics



The SETI Post-Detection Hub

What does humanity do when we discover we are not alone in the Cosmos?



Science fiction is awash with explorations of the impact on human society following discovery of, and even encounters with, life or intelligence elsewhere. But we need to go beyond thinking about the impact on humanity. We need to coordinate our expert knowledge not only for assessing the evidence but also for considering the human social response, as our understanding progresses and what we know and what we don't know is communicated. We have contemplated life on other Worlds for countless generations; the search for extra-terrestrial intelligence and their technology has now been an ongoing endeavour for decades. Our own technical resources have since improved significantly, and with those the chances of detection are therefore continually increasing. Social interest has risen considerably recently, due to the detection of thousands of planets around distant stars in our own galaxy and, closer to home, through the exploration of nearby worlds such as Mars and Venus, looking for signs of life, present or past. The potential discovery of microbial life will likely raise different types of concern that would follow the discovery of intelligent life – we are as yet entirely unprepared as a species for the latter. The time is thus right for consideration of humanity's response – and responsibility – following the detection of both life and intelligence in the Cosmos. We should plan now for this eventuality by setting out impact assessments, protocols, procedures and treaties designed to allow humanity to respond responsibly.

The SETI Post-Detection Hub is an initiative of the [UK SETI Research Network \(UKSRN\)](#), and is jointly hosted by the [St Andrews Centre for Exoplanet Science](#) and the [Centre for Global Law and Governance](#).



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End. Questions?